PHYS400
SPECIAL PROBLEMS IN PHYSICS
INTERIM REPORT

Project Title	Quantifying Information Extracted from Gravitational Wave Observations: An Empirical Test of the Flanagan–Hughes Framework Using GWTC Posterior Samples
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Field of study	Gravitational Wave Astrophysics
Type of study	Computational Data Analysis, Theoretical Testing
Submission Date	May 3, 2026

Introduction

The direct detection of gravitational waves (GWs) has opened a fundamentally new observational window into the universe (Abbott et al., 2016). The LIGO-Virgo-KAGRA (LVK) collaboration has completed four observing runs since the first observation of GW150914. The most recent data release, GWTC-4.0, brings the total number of compact binary merger candidates to 219 (Abbott et al., 2025). Each detected event includes physical information about its source. This includes component masses, spins, and sky location.

The extraction of this information is carried out using Bayesian parameter estimation, which is implemented through dedicated samplers such as those in LALInference and its successors (Veitch et al., 2015). The resulting posterior samples are made publicly available through the Gravitational Wave Open Science Center (LIGO Scientific Collaboration et al., 2023) and are read using the `pesummary` library (Hoy and Raymond, 2021). With next-generation detectors on the horizon, such as the Einstein Telescope and Cosmic Explorer, it has become increasingly important to understand the information-theoretic limits of these observations.

The amount of information extracted by a gravitational wave observation can be rigorously quantified using classical information theory. Shannon (1948) showed that the reduction in uncertainty produced by a measurement can be expressed in units of bits. In a Bayesian context, this quantity is known as the Kullback and Leibler (1951) (KL) divergence from the posterior to the prior distribution. The amount of information that can be extracted about a gravitational wave source's parameters is given by the following exact expression:

$$I_{\text{source}} = \int d\theta, p(\theta|s) \log_2 \left(\frac{p(\theta|s)}{p(\theta|p(s))} \right) \tag{1}$$

where $p(\theta|s)$ is the posterior distribution given the observed signal and $p(\theta|p(s))$ is the prior probability distribution given the magnitude of the measured signal $p(s)$. Computing this integral directly from real GWTC posterior samples is the central goal of this project.

Flanagan and Hughes (1998) introduced an analytical estimate of this quantity by assuming that detector noise is Gaussian and stationary, and waveform $h(\theta)$ depends on the source parameters (θ), is linear, and where there is little prior information. Under these conditions, **Equation 1** reduces to:

$$I_{\text{source}} \approx \frac{N_{\text{param}}}{2} \log_2 \left(1 + \frac{\rho^2}{N_{\text{param}}} \right) \tag{2}$$

where ρ is the matched-filter SNR and $N_{\text{param}} = 15$ for a binary black hole system, which covers the component masses, spin magnitudes and orientations, and extrinsic parameters. This approximation performs well in the high-SNR regime, but it becomes less reliable as SNR decreases. This is because non-Gaussian features in the posterior become more prominent. Despite its elegance, this framework has not been systematically tested against the actual posterior distributions produced by modern PE pipelines.

This project aims to address this gap. The central research question is: of all the gravitational wave events detected to date, how much information about source parameters has been obtained? We hypothesize that the real information that can be obtained will be close to, but not exactly the same as, the predictions made by Flanagan and Hughes. This is expected to be most noticeable for parameters that are strongly correlated or poorly constrained, particularly at low signal-to-noise ratio (SNR).

To test this hypothesis, we developed a Python-based pipeline using GWTC posterior samples to analyse a 15-dimensional parameter space. Non-parametric techniques such as k-nearest neighbours (k-NN) (Kraskov et al., 2004; Pérez-Cruz, 2008) are under investigation, but face significant challenges due to the curse of dimensionality. As an interim approach, we employ a multivariate Gaussian estimator (Muñoz and Ramele, 2026), which reduces the computational load at the cost of a distributional assumption that may not hold for all parameters.

Project Progress

Work Package 1: 1D Pipeline Verification

The initial phase of the project focused on developing and verifying the computational pipeline using a one-dimensional (1D) parameter model. This verification step was critical to ensure that our numerical algorithms for information extraction were functioning correctly before addressing the complexities of high-dimensional scaling. We chose the primary mass (m_1) of the GW150914 event as our primary test case. The information gain from the observation is quantified using the Kullback-Leibler (KL) Divergence, which measures the relative entropy between the posterior distribution $P(m_1|d)$ and the prior distribution $\pi(m_1)$:

$$D_{KL}(P||\pi) = \int P(m_1|d) \log_2 \left(\frac{P(m_1|d)}{\pi(m_1)} \right) dm_1 \tag{3}$$

To ensure the robustness of our results, we implemented two independent numerical approaches:

- Kernel Density Estimation (KDE): By approximating the continuous probability density from discrete posterior samples, we obtained a value of 2.638 bits.

- Binned Entropy Calculation: Applying the `scipy.entropy` method directly to binned samples yielded 2.6255 bits.

The high level of consistency between these results with an absolute difference of only ~ 0.01 bits—validates the accuracy of our integration and density estimation procedures.

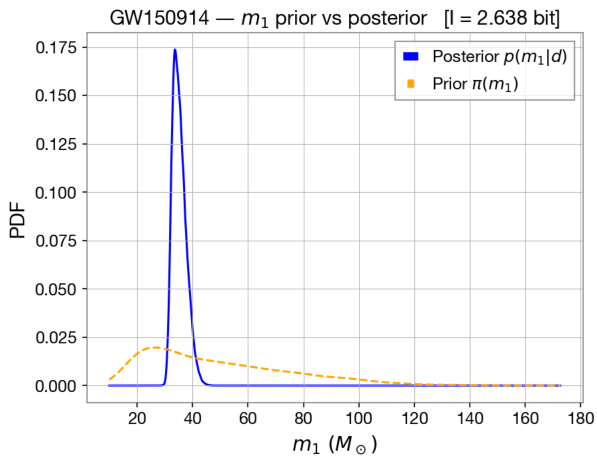


Figure 1: Comparison of prior and posterior distributions for the primary mass (m_1) of GW150914 via KDE. The information gain of 2.638 bits confirms the successful validation of the 1D computational pipeline.

As shown in Figure 1, the narrowing of the sharp posterior distribution relative to the broad prior visually confirms this information gain, representing a successful reduction in the system's entropy. This work package is complete, with no outstanding technical issues.

Work Package 2: 15-Dimensional Marginal KL Divergence

Following the calculation of the verified 1D parameters, we extended the calculation of the marginal KL divergence to the 15-parameter space of binary black hole mergers.

The parameters analyzed include the all primary physical properties, including the source-frame masses (m_1 [$M_\odot$], m_2 [$M_\odot$]), dimensionless spin magnitudes (a_1, a_2), spin tilt angles (tilt_1 [rad], tilt_2 [rad]), azimuthal angles (ϕ_{12} [rad], ϕ_{JL} [rad]), luminosity distance (d_L [Mpc]), inclination (θ_{JN} [rad]), polarization angle (ψ [rad]), sky position coordinates (azimuth [rad], zenith [rad]), geocentric time (t_c [s]), and the coalescence phase (phase [rad]). Independent 1D KDE was applied to each parameter, and the marginal KL divergences were summed to produce a total of 37.8915 bits, as shown in Figure 2.

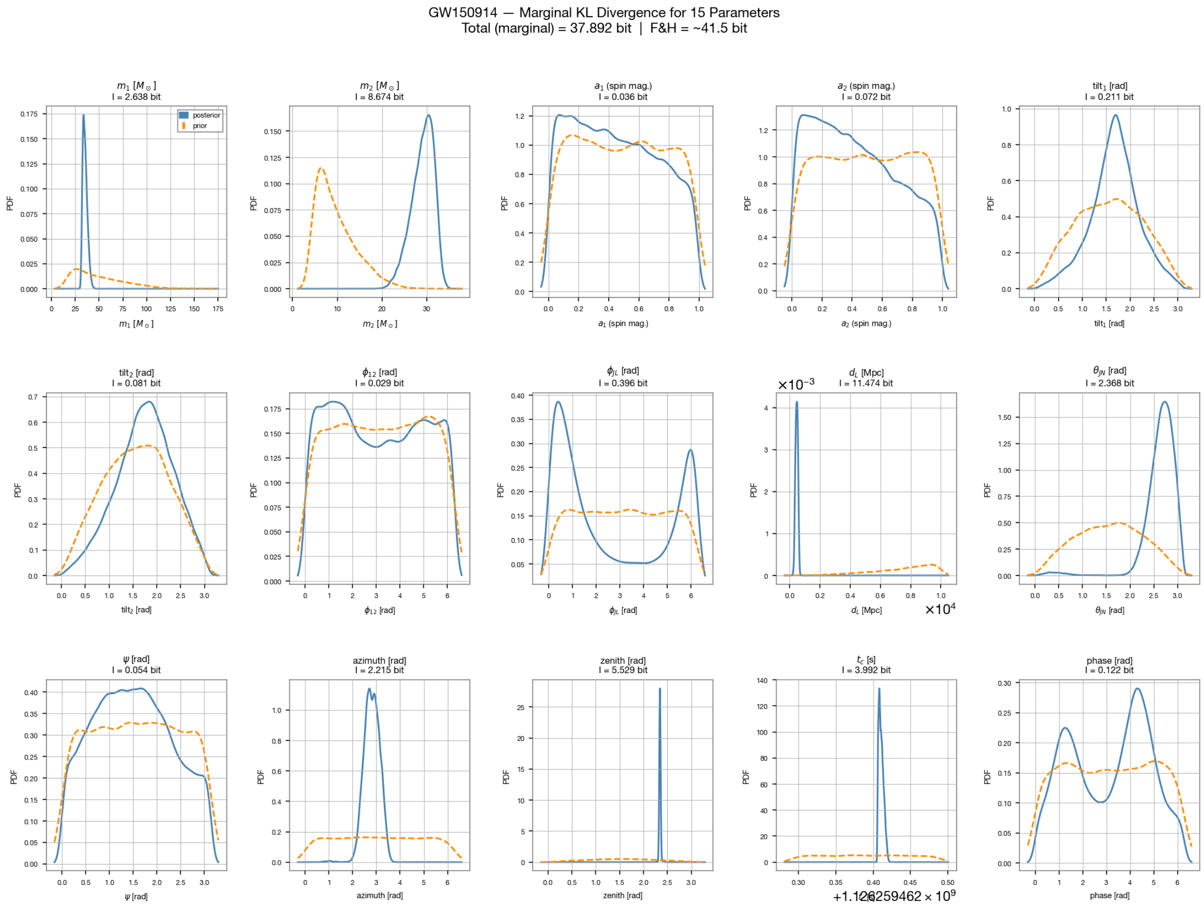


Figure 2: Marginal KL divergences for the 15 binary black hole parameters of GW150914, with a total marginal sum of 37.8915 bits. Observed non-Gaussian profiles in ϕ_{JL} and the coalescence phase indicate significant departures from Gaussianity.

This figure represents the sum of information stored in each marginal distribution. However, this additive approach encounters a significant limitation. This is because there is correlation between parameters. A simple summation of 1D KL divergences assumes implicitly that all 15 parameters are statistically independent. In reality, binary black hole observations show strong correlations and degeneracies. One example is the well-known coupling between luminosity distance (d_L) and orbital inclination (θ_{JN}).

Because this sum ignores parameter dependencies, the 37.8915-bit total is inherently incomplete. In order to accurately quantify the information gain, we must transition to high-dimensional joint-density estimation.

Work Package 3: Joint Density Estimation and the Flanagan-Hughes Framework

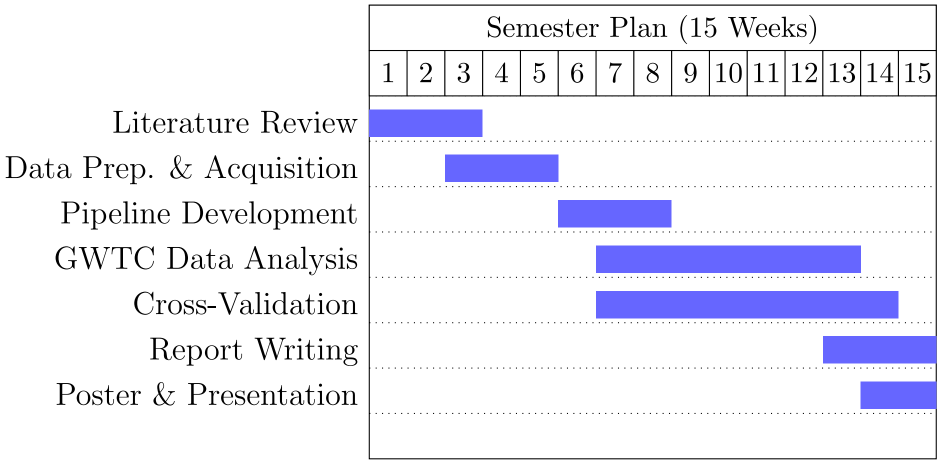
To perform direct joint density estimation for the 15-dimensional posterior, we initially attempted to utilize the `np.histogramdd` (Harris et al., 2020) function. However, this approach immediately encountered severe computational constraints, specifically the "Curse of Dimensionality (Altman and Krzywinski, 2018)" which resulted in a `MemoryError`.

Constructing a 15-dimensional grid, even with a minimal resolution of just 10 bins per dimension requires a total of 10^{15} bins. The amount of memory (RAM) needed to compute and store such a large grid is impossible to achieve with current hardware. Beyond these hardware limitations, we still have to address the critical 'sample sparsity' problem. When our approximately 147,000 posterior samples are distributed across 10^{15} bins, the vast majority of bins will be empty. This causes the local density to effectively drop to zero in nearly all regions of the parameter space. For these reasons, the use of `np.histogramdd` is entirely unusable given our specific parameter space and sample size.

To overcome these technical challenges, we switched to a framework that assumes a multivariate Gaussian distribution for the posterior samples. Using the analytical approach described in the referenced literature Muñoz and Ramele (2026), we calculated the information gain for the GW150914 event. Our empirical calculation yielded a total information content of **41.2466 bits**. When this result is compared to the theoretical calculation using the Flanagan–Hughes (FH) framework in Equation 2, with a parameter count ($\boldsymbol{N_{param} = 15}$) and a signal-to-noise ratio ($\boldsymbol{SNR = \rho = 26}$), the formula predicts a result of approximately **41.5** bits. The observed error margin of only **0.6%** indicates significant numerical alignment between our data-driven Gaussian baseline and the FH theoretical formula (Eqn 2).

Nevertheless, this numerical alignment must be interpreted with caution due to several limitations regarding our underlying assumptions. First, the multivariate Gaussian model is not universally valid for all parameters. As demonstrated in Figure 2, parameters such as the phase shift (ϕ_{JL}) and the coalescence phase often appear multimodal or nearly uniform, and are therefore inconsistent with a Gaussian distribution. Furthermore, the reliability of the FH framework itself is questionable, as the dependence on the source parameters, θ, is linear in the signal approximation used in its derivation, which may not be valid for real-world and high-SNR observations.

Finally, we explored the k-nearest neighbours (k-NN) Pérez-Cruz (2008) estimation method as a non-parametric alternative to overcome the limitations of bins. Although this method does not require any distributional assumptions, we have struggled to apply it to the 15D parameter space. Standard Euclidean distance metrics become poorly calibrated in high-dimensional spaces, which has made it challenging to obtain stable entropy estimates. Addressing these issues through parameter normalisation or adapted k-NN estimators remains our central computational challenge and will be the primary focus of the next phase.



Description of Work Packages

- **Literature Review:** Researching GW theory, Bayesian inference, and the Flanagan-Hughes theoretical framework.
- **Data Prep. & Acquisition:** Downloading GWOSC posterior samples and pre-processing them with the `pesummary` library.
- **Pipeline Development:** Implementing Python tools and conducting 1D verification using the m_1 parameter.
- **GWTC Data Analysis:** Calculating 15D marginal KL divergences and establishing the multivariate Gaussian.
- **Cross-Validation:** Comparing empirical results with FH predictions (achieving a 0.6% error margin) and troubleshooting k-NN scaling issues.
- **Report Writing:** Documenting numerical findings, technical bottlenecks, and progress for the interim and final PHYS 400 reports.
- **Poster & Presentation:** Synthesizing data visualizations and project conclusions for the final academic defense.

Future Outlook

The current method of measuring 15-dimensional joint KL divergence for a single gravitational wave event shows major limitations. While the multivariate Gaussian method provides a functional estimate, it may not accurately represent the physical reality of the posterior, as certain parameters have nonlinear behaviours and non-Gaussian distributions. To resolve this ambiguity, we will select two highly correlated parameters, such as the luminosity distance (d_L) and the orbital inclination (θ_{JN}), and calculate their joint KL divergence using both the Gaussian analytical approach and a non-parametric method. This will demonstrate whether the Gaussian approximation is sufficient for the full 15D case.

Relying on parametric assumptions alone is insufficient for a comprehensive analysis. Therefore, the project will prioritise developing non-parametric methods, with the k-nearest neighbours (k-NN) estimator as the primary candidate due to its ability to compute entropy without distributional assumptions. The work plan for the next one to two weeks involves troubleshooting the technical scaling issues currently encountered with the k-NN approach. Additionally, we will conduct a literature review and study on mutual information neural estimation (MINE) Belghazi et al. (2018) to determine its applicability as a neural-based solution for our quantification pipeline.

The final phase of the research will involve systematically comparing these diverse methodologies in order to evaluate the correctness of common assumptions, such as the assumption that the posterior is Gaussian. Once a validated method has been established for a single event, the process of quantifying information will be extended to the entire catalogue of binary systems. This comparative and multi-event analysis aims to provide a definitive empirical baseline for the physical information obtained from gravitational wave observations.

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